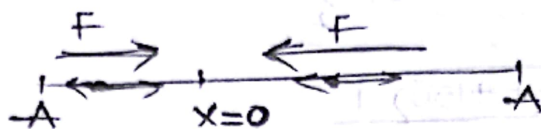


16 "SHM"

$$F = -Kx \quad \text{At } x=0 \quad (\because \text{Particle is in equilibrium})$$



- force acts always towards Mean Position...
- Mean Position, Extremum Positions...

$$F = -Kx$$

$$ma = -Kx$$

$$a = -\left(\frac{K}{m}\right)x$$

$$\text{let } \sqrt{\frac{K}{m}} = \omega$$

$$\frac{K}{m} = \omega^2$$

$$\textcircled{1} \quad a = -\omega^2 x$$

$$V \frac{dV}{dt} = -\omega^2 x$$

$$\textcircled{2} \quad V = \omega \sqrt{A^2 - x^2}$$

$$\frac{dx}{dt} = \omega \sqrt{A^2 - x^2}$$

$$\textcircled{3} \quad x = A \sin(\omega t + \phi)$$

displacement ~~into~~ Amplitude

$$\textcircled{4} \quad V = A\omega \cos(\omega t + \phi)$$

$$\textcircled{5} \quad a = -A\omega^2 \sin(\omega t + \phi)$$

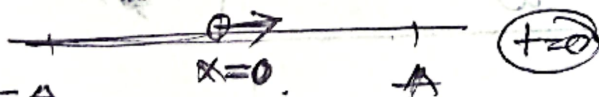
$$x = A \sin(\omega t + \phi)$$

Phase Angle.

ϕ : Initial Phase.

Case ①

$\phi = 0$. Mean Position moving towards Right Extremum.



Case ②

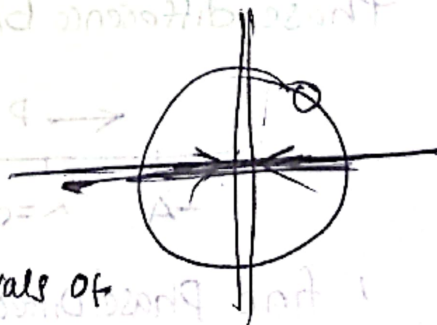
$\phi = 90$

NOTE

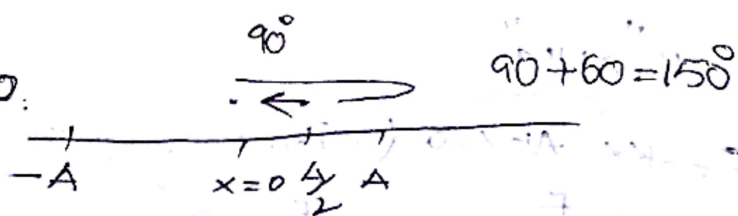
SHM Particle covers Equal

Phase Angles in Equal Intervals of

Time ...

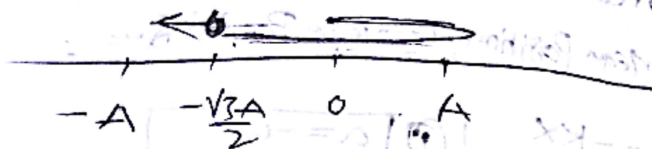


10. $t=0$



$$x = A \sin(\omega t + 150^\circ)$$

20.

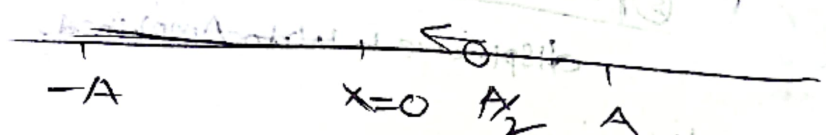


$$x = A \sin(\omega t + 240^\circ)$$

30.

$$x = A \cos(\omega t + \pi/3)$$

find min time taken by the particle to reach the Left Extreme Position

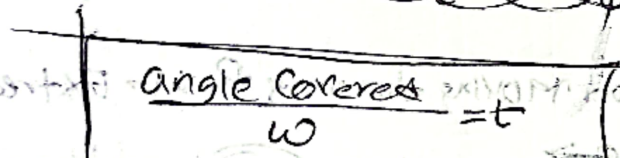


$$x = A \cos(\omega t + \pi/3)$$

$$x = A \sin(\omega t + \pi/2 + \pi/3)$$

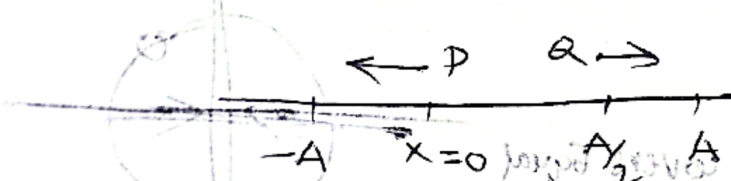
$$x = A \sin(\omega t + \frac{5\pi}{6})$$

$$\phi = \frac{5\pi}{6}$$



40.

Phase difference b/w Two particles: $\phi = \phi_1 - \phi_2$

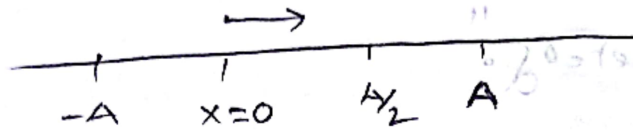


1. find Phase Difference b/w P & Q. P leads Q by ϕ'

$$\phi_1 - \phi_2 = \phi'$$

$$\pi - \frac{\pi}{6} = \phi' = 150^\circ$$

2. find the Position of Q at the Instant when P reached Left Extreme Position : $\frac{\sqrt{3}A}{2}$
3. ϕ' (Phase Difference b/w P and Q) will remain Unchange...
If Time Period of Particle is T.



$$360 \rightarrow 2\pi \quad T$$

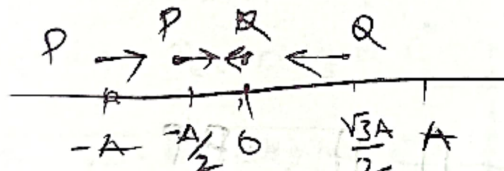
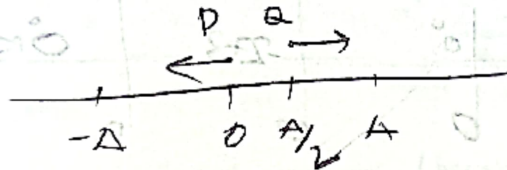
$$180 \rightarrow \pi \quad \frac{T}{2}$$

$$90 \rightarrow \frac{\pi}{2} \quad \frac{T}{4}$$

$$45 \rightarrow \frac{\pi}{4} \quad \frac{T}{8}$$

$$30 \rightarrow \frac{\pi}{6} \quad \frac{T}{12}$$

4. find the time at ~~which~~ which P & Q Cross Each Other...



$$185$$

$$165$$

$$t = \frac{185}{360} T = \frac{33T}{72}$$

Q

$$x = 3 + 48 \sin(\omega t + 150^\circ)$$

$$(x=3) = 48 \sin(\omega t + 150^\circ)$$

Extrem Mean Extrem

$$x=1 \quad x=3 \quad x=7$$

Q

$$x = 3 + 48 \sin \omega t$$

$$x = 3 + 2(2 - \cos 2\omega t)$$

$$x = 5 = -2 \cos(2\omega t)$$

$$x = 5 = 2 \sin 2\omega t + 3$$

Mean $x=8$
 $A=20$

find the position of the instant when it is released

$$U = \frac{1}{2} m \omega^2 A^2$$

$$KE = \frac{1}{2} m \omega^2 (A^2 - x^2)$$

Time period of particle is T